

DOOB’S PROOF OF MLE CONSISTENCY

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ABSTRACT. A short presentation of Doob’s proof of the consistency of Fisher’s maximum likelihood estimator.

INTRODUCTION

The consistency of Fisher’s maximum likelihood estimator (MLE) [10] is a basic result discovered a century ago. This result is typically subsumed under the rubric of M -estimators, and derived via standard techniques dating back to Cramér [6] and Wald [23]. For a more recent treatment, van der Vaart [22] is standard.

In teaching this material, I found Doob’s consistency proof [9] for the MLE more motivated and accessible to students, especially to students already exposed to information and entropy as part of a data science or machine learning course. The fact that the proof covers general parameter θ spaces and general state X spaces with no extra effort only adds to its appeal.

The aim of this note is to present this proof, together with some historical terminology remarks, in a self-contained manner. There is no novelty here except, perhaps, the conciseness of the presentation. By using the law of large numbers in its strong form, we derive MLE consistency in its strong form.

We start with the setup and assumptions, then state and prove consistency. After that, we attempt to give a historical account of entropy terminology, which is an interesting story in itself, then we end by showing the assumptions hold for the multivariate normal case.

SURPRISAL AND INFORMATION

Suppose a random variable X is sampled repeatedly, resulting in independent and identically distributed copies $X_1, X_2, \dots$ of X . We suspect the density of X is $p(X, \theta^*)$, one of a parametric family $p(X, \theta)$, and our goal is to recover θ^* from $X_1, X_2, \dots$.

Stated in this manner, all we have to work with are two items: the underlying expectation E_* and the family $p(X, \theta)$. Two issues immediately spring to mind. First, since we haven’t defined “density”, how is θ^* characterized? Second, once θ^* is well-defined, how do we use $X_1, X_2, \dots$ to approximate θ^* ?

To this end, we define the *surprisal* [17, 20]

$$(1) \quad S(X, \theta) = \log(1/p(X, \theta)).$$

This terminology is intuitive: low probability equals high surprisal. Then it is natural to characterize θ^* as a minimizer of the expected surprisal $E_*(S(X, \theta))$. More precisely, we say θ^* is a *true parameter* if θ^* is a minimizer of the expected surprisal $E_*(S(X, \theta))$, and the density corresponding to any minimizer is unique: $E_*(S(X, \theta)) = E_*(S(X, \theta^*))$ implies $p(X, \theta) = p(X, \theta^*)$.

Equivalently, let the *relative information* be

$$(2) \quad I(\theta^*, \theta) = E_*(S(X, \theta) - S(X, \theta^*)).$$

Then θ^* is a true parameter iff $I(\theta^*, \theta) \geq 0$ and $I(\theta^*, \theta) = 0$ only if $p(X, \theta) = p(X, \theta^*)$. The quantity I appears in many fields, in different forms, and is often called “relative entropy” or “Kullback-Leibler divergence”. Because of the widely differing terminology, we give below an account of the terminology history of I .

We say $p(X, \theta)$ is a parametric family of *densities* if there is¹ a background expectation E such that

$$E_\theta(f(X)) = E(p(X, \theta)f(X))$$

is an expectation for all θ . If $p(X, \theta)$ is a parametric family of densities and $E_{\theta^*} = E_*$, the relative information has the familiar form

$$I(\theta^*, \theta) = E \left(p(X, \theta^*) \log \left(\frac{p(X, \theta^*)}{p(X, \theta)} \right) \right),$$

and θ^* is² a true parameter [7, 8].

The simplest example is a categorical random variable X distributed according to $\theta^* = (\theta_1^*, \theta_2^*, \dots, \theta_d^*)$. If the values of X are $1, 2, \dots, d$, the mass function of X is $p(X, \theta^*) = \theta_X^*$, so $P_*(X = i) = \theta_i^*$, $i = 1, 2, \dots, d$. If $\theta = (\theta_1, \theta_2, \dots, \theta_d)$ is any other distribution, the surprisal is $S(X, \theta) = \log(1/\theta_X)$, and

$$(3) \quad I(\theta^*, \theta) = \sum_{i=1}^d \theta_i^* \log(\theta_i^*/\theta_i).$$

With this $I(\theta^*, \theta)$, the true parameter is θ^* .

The *likelihood* and *mean surprisal* are

$$L_n(\theta) = \prod_{k=1}^n p(X_k, \theta), \quad \bar{S}_n(\theta) = \frac{1}{n} \log(1/L_n(\theta)) = \frac{1}{n} \sum_{k=1}^n S(X_k, \theta).$$

The mean surprisal is also called the negative-log-likelihood.

By definition, θ^* minimizes $E_*(S(X, \theta))$. By the law of large numbers, $\bar{S}_n(\theta)$ converges to $E_*(S(X, \theta))$ as $n \rightarrow \infty$, with probability one. Hence it is natural to estimate θ^* by a minimizer of $\bar{S}_n(\theta)$ or, equivalently, a maximizer of $L_n(\theta)$: For each $n \geq 1$, a *maximum likelihood estimator (MLE)*, if it exists, is a parameter $\hat{\theta}_n$, depending on the outcome, maximizing $L_n(\theta)$ over all θ , for all outcomes. Under **B** below, with probability one, an MLE sequence $\hat{\theta}_n$ exists for all but finitely many n .

CONSISTENCY

To achieve consistency, there are three natural assumptions, the first of which is

A. (*Identifiability*): $\theta \neq \theta'$ implies $p(X, \theta) \neq p(X, \theta')$, hence $I(\theta^*, \theta) > 0$. Without identifiability, one cannot hope to extract θ from $p(X, \theta)$.

A function $F(\theta)$ is *proper* if its sublevel sets $F(\theta) \leq c$ are compact subsets of the parameter space, for every level c . A well-known elementary fact is the existence of minimizers of continuous proper functions. Since $\hat{\theta}_n$ should minimize $\bar{S}_n(\theta)$, this leads us to the second assumption

¹While $E_\theta(1) = 1$, E need not satisfy $E(1) = 1$, $E(1) = \infty$ is allowed.

²Apply Jensen’s inequality with the strictly convex $\log(1/p)$.

B. (Properness): $S(X, \theta)$ is integrable for all θ and there is $N \geq 1$, depending on the outcome, and a continuous proper $F(\theta)$, such that, with probability one,

$$(4) \quad F(\theta) \leq \bar{S}_n(\theta), \quad \text{for all } \theta \text{ and } n \geq N.$$

As a consequence of **B**, with probability one, an MLE $\hat{\theta}_n$ exists for $n \geq N$, and $I(\theta^*, \theta)$ is finite and proper.

We will need the convergence of $\bar{S}_n(\theta)$ to $E_*(S(X, \theta))$ across θ , in the sense

$$(5) \quad \bar{S}_n(\theta) \rightarrow E_*(S(X, \theta)), \quad \text{as } n \rightarrow \infty, \quad \text{for all } \theta,$$

with probability one. For this we need the third assumption

C. (Continuity): The parameter space is separable and there is a continuous $f(\theta, \theta')$ and an integrable $g(X)$ satisfying

$$|S(X, \theta) - S(X, \theta')| \leq f(\theta, \theta')g(X) \quad \text{and} \quad f(\theta, \theta) = 0.$$

Then a separability argument [18] establishes (5) with probability one.

B, **C** are typically checkable for *exponential families* [5, 14, 18, 22], the densities $p(X, \theta)$ where $S(X, \theta)$ is a sum of products of the form $f(\theta)g(X)$.

Theorem (Consistency). *Assume a population has a distribution depending on a parameter θ , and let θ^* be the population's true parameter. Then, under **A**, **B**, **C**, with probability one, an MLE sequence $\hat{\theta}_n$ may be defined for all but finitely many n , and any such sequence converges to θ^* as $n \rightarrow \infty$.*

In short, the known $\hat{\theta}_n$ is an approximation to the unknown θ^* .

Proof. By the law of large numbers,

$$(6) \quad \frac{1}{n} \sum_{k=1}^n g(X_k) \rightarrow E_*(g(X)), \quad \text{as } n \rightarrow \infty,$$

with probability one. Fix an outcome for which (4), (5), (6) hold, except, for (4), for finitely many n . We show $\hat{\theta}_n \rightarrow \theta^*$ as $n \rightarrow \infty$ for this outcome.

By (4),

$$F(\hat{\theta}_n) \leq \bar{S}_n(\hat{\theta}_n) \leq \bar{S}_n(\theta^*), \quad \text{for } n \geq N.$$

Since the right side is bounded as $n \rightarrow \infty$, and $F(\theta)$ is proper, the sequence $\hat{\theta}_n$ lies in a compact set in the parameter space.

Now suppose $\hat{\theta}_n \rightarrow \theta$ along a subsequence. By **C**,

$$(7) \quad \bar{S}_n(\theta) - \bar{S}_n(\theta^*) \leq \bar{S}_n(\theta) - \bar{S}_n(\hat{\theta}_n) \leq f(\theta, \hat{\theta}_n) \cdot \frac{1}{n} \sum_{k=1}^n g(X_k).$$

Since the left side of (6) converges to $E_*(g(X))$, and $f(\theta, \hat{\theta}_n) \rightarrow 0$ along a subsequence, the right side of (7) goes to zero along a subsequence. By (2) and (5), the left side of (7) converges to $I(\theta^*, \theta)$, hence $I(\theta^*, \theta) \leq 0$. But $I(\theta^*, \theta) \geq 0$, hence $I(\theta^*, \theta) = 0$. From **A**, this implies $\theta = \theta^*$. $\square$

TERMINOLOGY HISTORY

The relative information $I(\theta^*, \theta)$ is often called “relative entropy” [5, 14]. The term “entropy”, originating in thermodynamics, was coined by Clausius in his 1865 paper [4]. Boltzmann, in his 1872 paper [2], derived his H theorem, using a quantity H proportional to the negative of entropy. Gibbs, in his 1902 book [11], building on Boltzmann’s 1877 combinatorial interpretation [3] and Planck’s 1901 paper [16], flipped Boltzmann’s sign convention, writing the entropy H as done today.

Shannon, in his 1948 paper [19], independently defined

$$H(p) = - \sum_{i=1}^d p_i \log p_i.$$

Von Neumann had earlier defined quantum entropy in his 1932 book [15], and when Shannon consulted him around 1939–1940 about terminology for H , von Neumann suggested “entropy”, which Shannon adopted [21]. However, in his remark “ H is then, for example, the H in Boltzmann’s H theorem”, Shannon conflates his H with Boltzmann’s H .

The relative information $I(\theta^*, \theta)$ is jointly convex in the densities $p(X, \theta^*)$ and $p(X, \theta)$, a fact that is apparent in the special case (3), whereas $H(p)$ above is strictly concave in p . Because of this, calling both quantities “entropy” creates a mismatch. Indeed, based on the curvature, it is natural to call the *negative* of (3) the *relative entropy*. The mismatch is also in Python: `entropy(p)` returns the concave H , but `entropy(p, q)` returns the convex I .

This confusion can be dissolved by remembering entropy and information are opposites: entropy is concave and information is convex. More generally, this error is a common psychological conflation: When measuring the temperature, are we measuring how cold it is, or how hot it is? Of course we are doing both, but this doesn’t mean cold equals hot. In fact, cold equals minus hot.

While many authors’ terminology conflicts with ours, some authors align partially with our usage. Schervish, in his 1995 text [18], uses “Kullback-Leibler information” for I , which is more descriptive than the widely used but opaque “Kullback-Leibler divergence” [1]. Kullback himself, in his 1959 monograph [13], refers to I as an “information function”. Finally, Donsker-Varadhan, in their 1975 large deviations paper [8], denote the rate function as I rather than H , derive its properties in full generality, yet give it no name.

Regarding Doob’s 1934 proof, this was at the dawn of abstract probability theory, within the milieu of Birkhoff, Daniell, Fréchet, Hopf, Khintchine, Kolmogorov, Lévy, Radon, and Wiener, and within a year of Kolmogorov’s 1933 foundational extension theorem [12].

MULTIVARIATE NORMAL

Let μ be in $\mathbf{R}^d$, Q a positive definite symmetric $d \times d$ matrix, and $\theta = (\mu, Q)$. Let $v \cdot w$ be the dot product in $\mathbf{R}^d$. With x in $\mathbf{R}^d$, the *normal density with mean μ and variance Q* is $p(x, \theta) = e^{-S(x, \theta)}$ where

$$(8) \quad S(x, \theta) = \frac{1}{2} \log \det(2\pi Q) + \frac{1}{2} (x - \mu) \cdot Q^{-1} (x - \mu).$$

A random variable X in $\mathbf{R}^d$ is *normally distributed with parameter* $\theta^* = (\mu^*, Q^*)$ if its moment-generating function is

$$(9) \quad E_*(e^{v \cdot X}) = \exp \left(v \cdot \mu^* + \frac{1}{2} v \cdot Q^* v \right).$$

Suppose X is normally distributed with parameter $\theta^* = (\mu^*, Q^*)$ as in (9), and let $X_1, X_2, \dots$ be independent and identically distributed copies³ of X . We verify **A**, **B**, **C** for this setting. **A** and **C**, with $g(X) = 1 + |X|^2$, follow from (8). For **B**, we begin by reviewing background matrix results.

Given vectors v, w in $\mathbf{R}^d$, the *tensor product* $v \otimes w$ is the matrix whose ij -th entry is $v_i w_j$. Then $v \otimes v$ is symmetric,

$$(10) \quad v \cdot Qv = \text{trace}(Q(v \otimes v)), \quad v \otimes w + w \otimes v \leq \epsilon v \otimes v + \frac{1}{\epsilon} w \otimes w,$$

and, by the eigenvalue decomposition, every symmetric matrix is a linear combination of matrices of the form $v \otimes v$.

Given symmetric matrices Q, R, S , we say $Q \geq 0$ if $v \cdot Qv \geq 0$ for all v , and $Q \geq R$ if $Q - R \geq 0$. If $Q \geq 0$ and $R \geq 0$, then $\text{trace}(QR) \geq 0$. If $Q \geq 0$ and $R \geq S$, then $\text{trace}(QR) \geq \text{trace}(QS)$.

The determinant $\det(Q)$ is the product of the eigenvalues of Q . With Q invertible symmetric and t a scalar,

$$\det(Q + tv \otimes v) = \det(Q) (1 + tv \cdot Q^{-1}v).$$

If further $Q + tv \otimes v$ is invertible, then

$$(Q + tv \otimes v)^{-1} = Q^{-1} - \frac{t}{1 + tv \cdot Q^{-1}v} \cdot (Q^{-1}v) \otimes (Q^{-1}v).$$

These are the matrix facts used below.

Define $f(\mu) = Q^* + (\mu^* - \mu) \otimes (\mu^* - \mu)$ and

$$(11) \quad F(\mu, Q) = \frac{1}{2} \log \det(2\pi Q) + \frac{1}{2} \text{trace}(Q^{-1}f(\mu)).$$

In (9), multiply both sides by $e^{-v \cdot \mu}$, then replace v by tv and differentiate twice at $t = 0$. This yields

$$(12) \quad E_*(X) = \mu^*, \quad E_*((X - \mu) \otimes (X - \mu)) = f(\mu).$$

By (12), $S(X, \theta)$ is integrable, $E_*(S(X, \theta)) = F(\mu, Q)$, and

$$I(\theta^*, \theta) = E_*(S(X, \theta) - S(X, \theta^*)) = F(\mu, Q) - F(\mu^*, Q^*).$$

If $h(\lambda) = \log \lambda + 1/\lambda$ and $\lambda_1, \lambda_2, \dots, \lambda_d$ are the eigenvalues of $Z = QQ^{*-1}$, then $h(\lambda) \geq 1$ and $h(\lambda) = 1$ only if $\lambda = 1$, and

$$2F(\mu, Q) - 2F(\mu^*, Q^*) \geq \log \det(Z) + \text{trace}(Z^{-1} - I) = \sum_{i=1}^d (h(\lambda_i) - 1) \geq 0.$$

Hence θ^* is the true parameter.

Since $\text{trace}(f(\mu)) = \text{trace}(Q^*) + |\mu^* - \mu|^2$, $\text{trace}(f(\mu))$ is proper. Moreover, $f(\mu) \geq \delta I$ for some $\delta > 0$.

Lemma. *Let $f(\mu)$ be a symmetric matrix-valued function such that $\text{trace}(f(\mu))$ is proper and $f(\mu) \geq \delta I$ for some $\delta > 0$. Then $F(\mu, Q)$ in (11) is proper.*

³This setup is a special case of the Kolmogorov extension theorem.

Proof. Since $F(\mu, Q)$ is proper iff $F(\mu, \delta Q)$ is proper, and, up to a constant, $F(\mu, \delta Q)$ corresponds to $f(\mu)/\delta$, we may assume $\delta = 1$. Now suppose $2F(\mu, Q) \leq c$. It is enough to establish $\epsilon I \leq Q \leq I/\epsilon$ for some ϵ , since this confines Q to a compact subset of the space of positive definite matrices, and implies

$$c \geq 2F(\mu, Q) \geq \log \det(2\pi\epsilon I) + \epsilon \operatorname{trace}(f(\mu)),$$

which confines μ to a compact subset of $\mathbf{R}^d$.

If $h(\lambda) \leq c$, then $\lambda \leq e^c$. Since $\lambda \log \lambda \geq -1/e$ and $(e-1)e^c \geq ec$ for $c \geq 1$, $h(\lambda) \leq c$ and $c \geq 1$ implies $\lambda \geq (1-1/e)/c \geq e^{-c}$. Summarizing, with $\epsilon = e^{-c}$, $h(\lambda) \leq c$ implies $\epsilon \leq \lambda \leq 1/\epsilon$.

Let $\lambda_1, \lambda_2, \dots, \lambda_d$ be the eigenvalues of Q . Since $f(\mu) \geq I$,

$$(13) \quad \sum_{i=1}^d h(\lambda_i) = \log \det(Q) + \operatorname{trace}(Q^{-1}) \leq 2F(\mu, Q) - \log \det(2\pi I) \leq c.$$

Since $h(\lambda) \geq 1$, $h(\lambda_i) \leq c$, hence $\epsilon \leq \lambda_i \leq 1/\epsilon$, establishing $\epsilon I \leq Q \leq I/\epsilon$. $\square$

By the Lemma, $F(\mu, Q)$ in (11) is proper, hence $E_*(S(X, \theta))$ is proper, hence $I(\theta^*, \theta)$ is proper.

Let $0 < \epsilon < 1/2$ and let $\bar{X}_n = (X_1 + \dots + X_n)/n$ be the sample mean. By the law of large numbers and (12) with $\mu = \mu^*$, there is $N \geq 1$, depending on the outcome, such that, with probability one,

$$\frac{1}{n} \sum_{k=1}^n (X_k - \mu^*) \otimes (X_k - \mu^*) \geq (1-\epsilon)Q^*, \quad (\mu^* - \bar{X}_n) \otimes (\mu^* - \bar{X}_n) \leq \epsilon^2 Q^*, \quad n \geq N.$$

Let

$$f_\epsilon(\mu) = (1-2\epsilon)Q^* + (1-\epsilon)(\mu^* - \mu) \otimes (\mu^* - \mu).$$

Writing $X_k - \mu = (X_k - \mu^*) + (\mu^* - \mu)$ and using (10), with probability one,

$$\frac{1}{n} \sum_{k=1}^n (X_k - \mu) \otimes (X_k - \mu) \geq f_\epsilon(\mu), \quad \text{for all } \mu \text{ and } n \geq N.$$

Then, as before,

$$F_\epsilon(\theta) = F_\epsilon(\mu, Q) = \frac{1}{2} \log \det(2\pi Q) + \frac{1}{2} \operatorname{trace}(Q^{-1} f_\epsilon(\mu))$$

is proper, and, with probability one,

$$F_\epsilon(\theta) \leq \bar{S}_n(\theta), \quad \text{for all } \theta \text{ and } n \geq N,$$

establishing **B**.

By differentiating $\bar{S}_n(\mu + t\nu, Q)$ and $\bar{S}_n(\mu, Q + t\nu \otimes \nu)$ at $t = 0$, the MLE is

$$\hat{\mu}_n = \frac{1}{n} \sum_{k=1}^n X_k, \quad \hat{Q}_n = \frac{1}{n} \sum_{k=1}^n (X_k - \hat{\mu}_n) \otimes (X_k - \hat{\mu}_n).$$

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